

Shubham Sinha

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[LinkedIn](#), [GitHub](#), [LeetCode](#), [Intro video](#)

Summary

Software Engineer with 5+ years building scalable backend systems, search infrastructure, and applied AI with Python and Django. Scaled analytics on a billion-row database by **50%** and cut search index latency from hours to under a minute. Strong in REST APIs, microservices, event-driven architecture, and AWS.

Work Experience

Numerator

Jan 2024 – Present

Software Engineer I

Remote, India

- Optimized reporting queries on a billion-row MySQL database with the data engineering team, cutting heavy report runtimes from minutes to seconds and improving throughput by **50%**.
- Re-architected a Django REST service over Elasticsearch into decoupled business API, search, and ingestion layers, reducing median API response time by **150ms**.
- Replaced nightly search-index rebuilds with MySQL change data capture using pub/sub and Celery, cutting indexing latency from overnight to under **60 seconds**.
- Automated AWS infrastructure (EC2, RDS, S3, Lambda) with Terraform, reducing new environment provisioning from hours to minutes.
- Fine-tuned Hugging Face models and spaCy text pipelines for US, Canada, and Germany catalogs, improving brand and category matching accuracy by **23%**.

Inexture Solutions LLP

Dec 2020 – Dec 2023

Python Developer

Ahmedabad, Gujarat, India

- Built UPI deposit flows for a crypto wallet serving **19M+** registered users, enabling card-free wallet funding.
- Developed Grafana and ELK dashboards for payment systems, surfacing issues earlier and reducing mean time to recovery (MTTR) by **40%**.
- Decoupled services for a leading job portal using RabbitMQ, isolating slow services to improve system reliability.
- Automated multi-stage builds and releases with Jenkins CI/CD, eliminating most manual deployment steps.

Technical Skills

Languages: Python, SQL, Go

Backend & Data: Django, Django REST Framework, FastAPI, REST APIs, Microservices, MySQL, PostgreSQL, MongoDB, Elasticsearch, Celery, RabbitMQ, Kafka, Debezium, SQLAlchemy

Cloud & DevOps: AWS, Terraform, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Git, Linux, Grafana, Prometheus, ELK

AI / NLP: LangGraph, LangChain, Hugging Face Transformers, spaCy, RAG, Vector Search, System Design, Event-Driven Architecture

Projects

Incident Commander

Python, FastAPI, LangGraph, LangChain, PostgreSQL, SSE

- Built an on-call assistant with LangGraph where **4** agents investigate logs, metrics, recent changes, and runbooks, and a planner synthesizes findings into a diagnosis before any risky action.
- Enforced human-in-the-loop approval before risky steps, persisted progress in Postgres checkpoints to resume runs without re-investigation, and streamed live updates over SSE from PagerDuty and Datadog webhooks.

PulseSearch

Python, FastAPI, Debezium, Kafka, Elasticsearch, RAG, Kubernetes

- Streamed MySQL changes through Debezium and Kafka into Elasticsearch, combining keyword and vector search for hybrid retrieval with RAG answers grounded only in retrieved results.
- Engineered idempotent, replay-safe indexing with version checks and a dead-letter queue, cutting search sync from a ~12-hour overnight run to under **60 seconds**; deployed on Kubernetes via GitHub Actions to GHCR.

Education

Dharmsinh Desai University

Bachelor of Technology in Information Technology (IT)

Nadiad, Gujarat, India

Aug 2017 – May 2021